$$T: \mathrm{Poly}_{\le 4}(F) \longrightarrow \mathrm{Poly}_{\le 4}(F)$$

$$f(x) \longmapsto x f'(x) + f(x)$$

Choose basis $(1, x, x^2, x^3, x^4)$ for both sides

$$T: \begin{array}{lcl} 1 & \longmapsto & 1 \\ x & \longmapsto & 2x \\ x^2 & \longmapsto & 3x^2 \\ x^3 & \longmapsto & 4x^3 \\ x^4 & \longmapsto & 5x^4 \end{array}$$

The matrix $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}$

As long as these inverses

$R2 \times 2^{-1}$

$R3 \times 3^{-1}$

$R4 \times 4^{-1}$

exist $\downarrow R_5 \times 5^{-1}$

$$\text{RREF of } A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= I_5$$

The only solution to $[A \mid 0] \Leftrightarrow [I_5 \mid 0]$

is $x_1 = x_2 = x_3 = x_4 = x_5 = 0$

$\Rightarrow$ $\ker T = \{0\}$

Moreover, $\operatorname{im} T = \mathrm{Poly}_{\le 4}(F)$

T is actually an isomorphism

Observation:

V_1 & V_2 : f. dim'l F-vector spaces of the same dimension.

$$T : V_1 \longrightarrow V_2$$

TFAE:

(1) T is an isomorphism

(2) T is injective $\iff \ker(T) = \{0\}$

(3) T is surjective $\iff \operatorname{im}(T) = V_2$

(4) For some choice of bases $\mathcal{B}_1$ for V_1 & $\mathcal{B}_2$ for V_2, the matrix $A = [T]_{\mathcal{B}_1, \mathcal{B}_2}$ is invertible

(5) The RREF of A is I_n where $n = \dim V_1 = \dim V_2$

Pf: ① $\Leftrightarrow$ ④ ✓, ① $\Rightarrow$ ② ✓

① $\Rightarrow$ ③ ✓

Fact: T is injective $\Leftrightarrow$ $\ker T = \{0\}$

Pf: If T is injective, then

($\Rightarrow$) $T(x) = 0$ will have 0 as a solution

($\Leftarrow$) If $\ker T = (0)$, we want to show:

$$T(v_1) = T(v_2) \Rightarrow v_1 = v_2$$

$$\Big\Downarrow -T(v_2) \qquad\qquad \Uparrow$$

$$v_1 - v_2 = 0$$

$$\Uparrow \leftarrow \ker T = \{0\}$$

$$T(v_1) - T(v_2) = 0 \Rightarrow T(v_1 - v_2) = 0$$

$\uparrow$

T is a linear transformation

(2) $\Leftrightarrow$ (3) :

Rank-nullity tells us

$$\underbrace{\dim \ker(T)}_{=0} + \underbrace{\dim \operatorname{im}(T)}_{=n} = n$$

$\underbrace{\dim \ker(T)}_{=0} \Leftrightarrow \underbrace{\dim \operatorname{im}(T)}_{=n}$

$n = \dim V_2 = \dim V_1 = n$

$\Downarrow$

$\operatorname{im}(T) = V_2 \Leftrightarrow \ker T = \{0\}$

(4) $\Leftrightarrow$ (5)

$[A \mid 0]$ only has 0 as a solution

$\Updownarrow$

$[\mathrm{RREF}(A) \mid 0]$ only has 0 as a solution

$\Updownarrow$

Every column of $\mathrm{RREF}(A)$ has a leading 1

$(\Leftarrow)$ $\mathrm{RREF}(A) = I_n$

Example: Let's now assume

$$F = \mathbb{Z}/2\mathbb{Z}$$

Then

$$A = \begin{array}{c} \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right] \\ \begin{array}{ccccc} e_1 & & e_3 & & e_5 \end{array} \end{array}$$

$$\Big\downarrow \begin{array}{l} R2 \Leftrightarrow R3 \\ R5 \Leftrightarrow R3 \end{array}$$

$$\mathrm{RREF}(A) = \begin{array}{c} \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \\ \begin{array}{ccccc} e_1 & & e_2 & & e_3 \end{array} \end{array}$$

$[A \mid 0]$: Solutions are

$$x_1 = x_3 = x_5 = 0$$

i.e. $\left\{ a\begin{bmatrix}0\\1\\0\\0\\0\end{bmatrix} + b\begin{bmatrix}0\\0\\0\\1\\0\end{bmatrix} : a,b \in F \right\}$

$$= \left\{ \begin{bmatrix}0\\a\\0\\b\\0\end{bmatrix} : a,b \in F \right\}$$

$\ker T = \{ ax + bx^3 : a,b \in F \}$

$\operatorname{im} T = \operatorname{Span}(1, x^2, x^4)$